

Comprehensive Customer Intelligence Report

SQL-Driven Retention Strategy

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1 Project Overview

This project focuses on analyzing customer transactional and behavioral data for a direct-to-consumer fashion brand. The objective of the analysis is to identify loyal customers, understand promotional dependency, evaluate customer retention behavior, and generate strategic recommendations for improving long-term customer value.

The project combines:

- Python-based feature engineering
- SQL-driven customer analytics
- Business intelligence interpretation
- Retention-focused recommendations

The final goal is to transform raw customer data into actionable business insights that support strategic decision-making.

2 Business Problem

The company currently relies heavily on promotional campaigns and discounts to drive sales. However, the business lacks visibility into:

- Which customers are genuinely loyal
- Which customers are highly dependent on discounts
- Which product categories drive retention
- Which regions demonstrate strong organic demand
- How customer satisfaction impacts loyalty

3 Dataset Description

The dataset contains customer-level and transactional information for a fashion e-commerce business.

3.1 Dataset Features

Column	Description
Customer ID	Unique customer identifier

Age	Customer age
Gender	Customer gender
Category	Product category purchased
Purchase Amount (USD)	Purchase value
Review Rating	Customer satisfaction rating
Discount Applied	Whether discount was used
Promo Code Used	Whether promo code was used
Previous Purchases	Historical purchase count
Season	Purchase season
Location	Customer region/location

The dataset was cleaned and transformed into a customer intelligence dataset through feature engineering and behavioral segmentation.

4 Feature Engineering

Feature engineering was performed to create business-relevant customer intelligence metrics.

4.1 Key Engineered Features

Feature	Business Purpose
Promo Dependency Score	Measures reliance on discounts/promotions
Loyalty Score	Measures long-term customer engagement
Customer Value Tier	Segments customers by spending level
Satisfied Customer Flag	Identifies satisfied users
At-Risk Customer Flag	Detects potentially disengaged customers
Loyal Customer Flag	Identifies high-retention customers

4.2 Loyal Customer Definition

Loyal customers were intentionally defined as customers exhibiting:

- High repeat purchase behavior
- Low promotional dependency

Specifically, customers with:

$$Promo_Dependency_Score < 0.5$$

and strong historical purchase behavior were classified as loyal customers.

5 SQL Business Analysis

This section contains SQL-driven customer intelligence analysis.

5.1 Insight 1: Customer Value Tier Analysis

SQL Query

```
SELECT
    Customer_Value_Tier ,
    COUNT(*) AS total_customers ,
    ROUND(AVG(Total_Spend),2) AS avg_spend
FROM feature_engineered_customers
GROUP BY Customer_Value_Tier;
```

Output

Value Tier	Total Customers	Avg Spend
High	1282	87.22
Medium	1288	59.73
Low	1293	32.38

Insight

High-value customers contribute significantly larger revenue compared to low-value customers.

5.2 Insight 2: Loyalty vs Promotional Dependency

SQL Query

```
SELECT
    Loyal_Customer ,
    COUNT(*) AS customers ,
    ROUND(AVG(Total_Spend),2) AS avg_spend ,
    ROUND(AVG(Promo_Dependency_Score),2)
    AS avg_promo_dependency
FROM feature_engineered_customers
GROUP BY Loyal_Customer;
```

Key Finding

Loyal customers demonstrate near-zero promotional dependency while maintaining comparable spending behavior.

Business Meaning

The company's most loyal customers are driven primarily by brand engagement rather than discounts.

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5.3 Insight 3: Satisfaction vs Loyalty Analysis

SQL Query

```
SELECT
    Satisfied_Customer ,
    ROUND(AVG(Loyalty_Score),2) AS avg_loyalty
FROM feature_engineered_customers
GROUP BY Satisfied_Customer;
```

Output

Satisfied Customer	Avg Loyalty Score
0	13.52
1	14.05

Insight

The analysis indicates a positive relationship between customer satisfaction and loyalty behavior. Customers with higher satisfaction levels tend to exhibit stronger repeat engagement and long-term retention characteristics.

Business Meaning

Customer retention is influenced not only by discounts and promotions, but also by customer experience and satisfaction levels.

This suggests that:

- Satisfied customers are more likely to remain engaged with the brand
- Positive customer experience contributes to long-term customer value
- Experience-driven retention can support sustainable revenue growth

The findings reinforce the importance of treating customer satisfaction as a strategic business metric rather than only an operational measure.

5.4 Insight 4: Category-Wise Retention Analysis

SQL Query

```
SELECT
    Category ,
    ROUND(AVG('Purchase Amount (USD)') ,2)
    AS avg_purchase ,
    ROUND(AVG('Previous Purchases ') ,2)
    AS avg_previous_purchases
FROM dataset
GROUP BY Category
ORDER BY avg_previous_purchases DESC;
```

Output

Category	Avg Purchase	Avg Previous Purchases
Accessories	59.79	25.68
Footwear	60.23	25.28
Clothing	59.93	25.18
Outerwear	57.15	25.00

Key Finding

Accessories and footwear categories demonstrate stronger repeat purchase behavior.

Insight

Certain product categories contribute more strongly to customer retention than others.

Business Meaning

Accessories and footwear appear to function as retention-driving categories, while outerwear behaves more like an occasional purchase category.

5.5 Insight 5: High Promotional Dependency Analysis

SQL Query

```
SELECT
    COUNT(*) AS high_promo_customers
FROM feature_engineered_customers
WHERE Promo_Dependency_Score > 0.7;
```

Output

High Promo Customers
1640

Key Finding

A large segment of customers demonstrates strong dependence on promotions.

Insight

The business currently relies heavily on discount-driven customer engagement.

Business Meaning

High promotional dependency may reduce profitability and create unstable long-term retention.

5.6 Insight 6: At-Risk Customer Analysis

SQL Query

```
SELECT
    At_Risk_Customer ,
    ROUND(AVG(Total_Spend),2) AS avg_spend
FROM feature_engineered_customers
GROUP BY At_Risk_Customer;
```

Output

At Risk Customer	Avg Spend
1	59.02
0	60.15

Key Finding

At-risk customers still contribute meaningful revenue to the business.

Insight

Potentially disengaged customers continue to hold commercial value and should not be ignored.

Business Meaning

Revenue loss can occur if early retention signals are not identified proactively.

5.7 Insight 7: Ideal Customer Profile Analysis

SQL Query

```
SELECT
    Gender ,
    Location ,
    Customer_Value_Tier ,
    ROUND(AVG(Loyalty_Score),2)
    AS avg_loyalty_score ,
    ROUND(AVG(Promo_Dependency_Score),2)
    AS avg_promo_dependency
FROM feature_engineered_customers
WHERE Loyal_Customer = 1
GROUP BY Gender, Location, Customer_Value_Tier
ORDER BY avg_loyalty_score DESC;
```

Output

High-value loyal customers from regions such as Ohio, California, and Oklahoma demonstrated the strongest loyalty behavior with near-zero promotional dependency.

Key Finding

High-value customers exhibit strong loyalty while maintaining minimal dependency on promotions.

Insight

The brand's strongest customers are organically engaged rather than discount-driven.

Business Meaning

These customer segments likely contribute higher long-term customer lifetime value and stronger retention stability.

5.8 Insight 8: Revenue Dependency on Promotions

SQL Query

```
SELECT
    ROUND (
        SUM (
            CASE
                WHEN Promo_Dependency_Score > 0.7
                THEN Total_Spend
                ELSE 0
            END
        ), 2
    ) AS promo_dependent_revenue
FROM feature_engineered_customers;
```

Output

Promo Dependent Revenue
96959

Key Finding

A substantial portion of total revenue depends on highly promotion-driven customers.

Insight

The company's current sales performance is strongly supported by discount-based purchasing behavior.

Business Meaning

Over-reliance on promotions may create long-term profitability and retention risks.

5.9 Insight 9: Seasonal Retention Analysis

SQL Query

```
SELECT
    Season ,
    ROUND(AVG('Previous Purchases'), 2)
    AS avg_previous_purchases ,
    ROUND(AVG('Purchase Amount (USD)'), 2)
    AS avg_purchase
```

```
FROM dataset
GROUP BY Season
ORDER BY avg_previous_purchases DESC;
```

Output

Season	Avg Previous Purchases	Avg Purchase
Winter	25.94	60.26
Summer	25.40	58.37
Spring	25.05	58.64
Fall	24.98	61.53

Key Finding

Winter demonstrates the strongest repeat purchase behavior while Fall generates the highest average purchase value.

Insight

Customer engagement and spending behavior vary across seasons.

Business Meaning

Seasonal behavioral patterns can influence retention and revenue performance.

5.10 Insight 10: Geographic Revenue Analysis

Objective

Identify geographic regions with strong spending behavior and low promotional dependency.

SQL Query

```
SELECT
    Location,
    COUNT(*) AS total_customers,
    ROUND(AVG(Total_Spend),2) AS avg_spend,
    ROUND(AVG(Promo_Dependency_Score),2)
    AS avg_promo_dependency
FROM feature_engineered_customers
GROUP BY Location
ORDER BY avg_spend DESC,
avg_promo_dependency ASC;
```

Output

Regions such as Alaska, Arizona, and Pennsylvania demonstrated strong average spending with relatively low promotional dependency.

Key Finding

Certain geographic regions show strong organic customer demand.

Insight

Customer behavior varies significantly across geographic regions.

Business Meaning

High-spend regions with low promotional dependency represent commercially attractive markets for long-term growth.

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6 Power BI Dashboard Analysis

An interactive Power BI dashboard was developed to visualize customer behavior, promotional dependency, retention trends, and geographic opportunities.

The dashboard includes four major analytical visuals:

- **Customer Revenue Pyramid** — shows customer distribution across high, medium, and low value tiers.
- **Geographic Opportunity Map** — identifies regions with high spending and relatively low promotional dependency, indicating stronger organic demand.
- **Promo Dependency vs Retention Analysis** — compares customer loyalty against promotional dependency to identify organically loyal and discount-driven customer segments.
- **Category Retention Funnel** — highlights product categories with stronger repeat purchase behavior and retention potential.

The dashboard also includes KPI cards such as:

- Total Revenue
- Total Customers
- Loyal Customers

- Promo-Dependent Revenue

Overall, the dashboard provides a concise executive-level overview of customer intelligence and supports retention-focused business decision making. “

7 Retention Playbook Summary

This section summarizes the strategic recommendations derived from the customer intelligence analysis. The objective of the playbook is to improve long-term customer retention while reducing excessive dependency on promotional sales.

7.1 Promotional Sunset Strategy

The analysis revealed that a substantial portion of customers demonstrate high promotional dependency, indicating strong reliance on discounts for purchasing behavior.

At the same time, loyal high-value customers exhibited:

- Strong repeat purchase behavior
- Higher average spending
- Low promotional dependency

Based on these findings, the company should gradually reduce blanket discount campaigns for organically loyal customers and transition toward:

- loyalty rewards
- personalized engagement
- premium membership benefits
- experience-driven retention strategies

This approach can help improve long-term profitability while maintaining customer retention stability.

7.2 Ideal Customer Profile

The analysis identified the company’s strongest customer segment as:

High-value customers from strong-performing geographic regions who demonstrate high repeat purchase behavior, low promotional dependency, and consistent spending patterns.

These customers represent the company’s most commercially valuable long-term customer base.

Additionally, categories such as:

- Accessories
- Footwear

demonstrated stronger retention behavior and may serve as important retention-driving categories.

7.3 Strategic Recommendations

Based on the analysis, the following strategic priorities are recommended:

1. Reduce excessive dependency on blanket promotional campaigns
2. Prioritize retention of high-value loyal customers
3. Focus on customer experience-driven engagement
4. Expand marketing efforts in organically strong regions
5. Monitor at-risk customers proactively

The overall strategic direction should shift from:

discount-driven growth

toward:

retention-driven sustainable growth

8 Conclusion

This project successfully transformed raw transactional and behavioral customer data into actionable customer intelligence insights through feature engineering, SQL-driven analysis, and business-focused dashboarding.

The analysis identified several important business patterns:

- Loyal customers demonstrated low promotional dependency and strong repeat purchase behavior
- A substantial portion of revenue remains promotion-driven, creating long-term profitability risk

- Customer satisfaction positively influences loyalty and retention
- Certain product categories and geographic regions demonstrate stronger organic retention potential
- High-value customers contribute disproportionately to overall business value

The project further translated analytical findings into strategic business recommendations through the retention playbook, including promotional sunset strategies and ideal customer profiling.

Overall, the findings suggest that the company should gradually transition from:

discount-driven growth

toward:

retention-driven sustainable growth

by prioritizing customer experience, loyalty-based engagement, and long-term customer value optimization.

9 Future Improvements

Future work can further strengthen the customer intelligence framework through more advanced analytical and predictive capabilities.

Potential improvements include:

- Machine learning-based churn prediction models
- Customer lifetime value (CLV) forecasting
- Advanced customer segmentation and clustering
- Personalized recommendation systems
- Time-series retention and seasonal forecasting analysis
- Real-time Power BI dashboards with automated data refresh
- Predictive promotional optimization models

Additionally, integrating real-world transactional streams and larger-scale customer datasets may further improve the accuracy and business applicability of the retention strategies proposed in this project.